

Set-2 Scenario Based

1.

- Check the label format and dataset annotations, other classes might be missing
- Ensure the number of classes is correctly passed to the model
- Dataset may be imbalanced, so that the model is biased toward one class.

2.

- Instead of fine tuning the entire model freeze early layers and train only the final classification layer
- Use the pre-trained weights as feature extractor

3.

- Perform domain specific fine tuning by augmenting or collecting night time images
- Apply brightness and contrast augmentation or use real images under different lighting

4.

- Ensure label files follow YOLO format:[class_id,x_center,y_center,width,height] in normalized form
- Match training image size and validation image size

5.

- Calibrate webcam feed, resize to same resolution as training.
- Collect some real time webcam samples and fine tune the model
- Apply real time test time augmentation techniques

6.

- VGG is computationally heavy. Try feature extraction mode by freezing all convolutional layers.
- Replace the fully connected layers with a lighter classifier

7.

- InceptionV3 expects input size of 299x299(not 224x224 like ResNet/VGG).

- Resize all training and validation images accordingly.